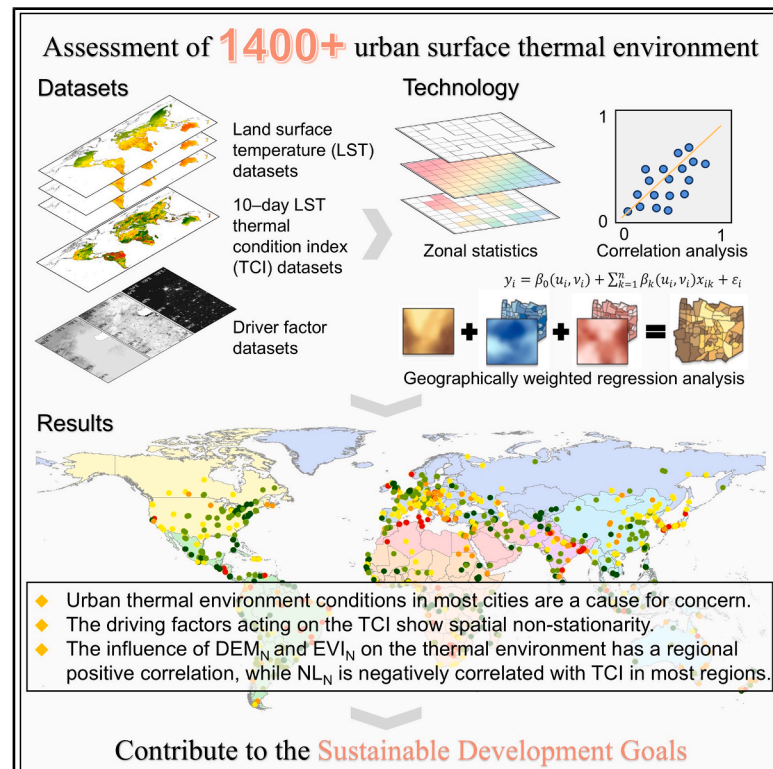


Continuous assessment of the factors driving the urban surface thermal environment in 1,469 cities worldwide

Graphical abstract



Authors

Jun Yang, Jiayi Ren, Felix Creutzig, ..., Xiangming Xiao, Jianhong (Cecilia) Xia, Quansheng Ge

Correspondence

yangjun8@mail.neu.edu.cn (J.Y.),
geqs@igsrr.ac.cn (Q.G.)

In brief

Copernicus TCI data are a more standardized version of the 10-day LST. A global estimate of urban surface thermal environment spatial heterogeneity and its driving factors is explored via quantitative remote sensing and GIS spatial analysis, enhancing the current understanding of the urban surface thermal environment on a global scale.

Highlights

- TCI data offer a more standardized perspective of the 10-day LST
- 53% of cities are facing heat stress that is higher than the global average
- NL_N and EVI_N are high in warm and cold areas, consistent with TCI in climate zones
- Vegetation cover is a weaker regulator than topographic and socioeconomic drivers are



Article

Continuous assessment of the factors driving the urban surface thermal environment in 1,469 cities worldwide

Jun Yang,^{1,2,9,10,11,*} Jiayi Ren,^{1,10} Felix Creutzig,³ Bingyu Zhao,^{3,4} Wei Sun,⁵ Xiangming Xiao,⁶ Jianhong (Cecilia) Xia,⁷ and Quansheng Ge^{8,*}

¹School of Humanities and Law, Northeastern University, Shenyang 110169, China

²Human Settlements Research Center, Liaoning Normal University, Dalian 116029, China

³Sustainability Economics of Human Settlements, Technische Universität Berlin, Berlin 10623, Germany

⁴Institute of Applied Ecology, Chinese Academy of Sciences, Shenyang 110016, China

⁵Nanjing Institute of Geography and Limnology, Chinese Academy of Sciences, Nanjing 210008, China

⁶Center for Earth Observation and Modeling, University of Oklahoma, Norman, OK 73019, USA

⁷School of Earth and Planetary Sciences (EPS), Curtin University, Perth, WA 65630, Australia

⁸Institute of Geographical Sciences and Natural Resources, Chinese Academy of Sciences, Beijing 100101, China

⁹Jangho Architecture College, Northeastern University, Shenyang 110169, China

¹⁰These authors contributed equally

¹¹Lead contact

*Correspondence: yangjun8@mail.neu.edu.cn (J.Y.), geqs@igsrr.ac.cn (Q.G.)

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SCIENCE FOR SOCIETY Global warming has increased the risk of urban heat, making it crucial to assess urban surface thermal environments and their adaptive responses. This study enhances the understanding of urban thermal dynamics by utilizing geostationary satellite data and identifies urban surface thermal environments related to topography, vegetation, and nighttime lighting in 1,469 cities worldwide. It is crucial to effectively understand existing natural and social resources and the state of the surface thermal environment in cities. Our study overcomes the limitations of individual cases and makes substantive progress in analyzing the surface thermal environment on a global scale, yielding statistically significant results. Different climate zones exhibit significant differences in abnormal thermal environment characteristics, with cities in arid (0.32) and hot (0.28) regions having lower average TCI values than those in warm (0.41) and cold (0.36) regions. We emphasize the need for thermal management in small and medium-sized cities in hot and arid regions. The results elucidate the differences in the surface thermal environment, promote inter-regional coordination and management in climate adaptation efforts, and support the global Sustainable Development Goals.

SUMMARY

Identifying global heterogeneity in surface thermal environments is crucial for urban cooling. The 10-day thermal condition index (TCI) from geostationary satellites represents the warmth of individual pixels (with a spatial resolution of 5 km) relative to their historical climatic temperature range, enhancing the understanding of local thermal conditions. Here, we report the global patterns and correlations of the TCI in 2021 in relation to topography (measured as DEM_N), vegetation cover (measured as EVI_N), and socioeconomic factors (measured as NL_N). The results show that 53% of cities are facing heat stress that is higher than the global average. The driving factors acting on the TCI show spatial non-stationarity. The influences of DEM_N and EVI_N on the TCI show positive regional correlations, while NL_N and TCI mainly show negative spatial correlations. From a global perspective, the vegetation regulation effect is relatively weak. The results provide support for the realization of the Sustainable Development Goals.

INTRODUCTION

Globally, climate warming and industrialization have led to substantial urban land surface heating,^{1–3} exacerbating the heat

risks associated with human survival and urban development, such as urban heat island effects and heat-related mortality.^{4–8}

Addressing heat-induced effects on human well-being is included in the United Nations Sustainable Development Goals



(SDGs).⁹ Specifically, addressing extreme heat in urban environments is directly related to SDG 3, which focuses on Good Health and Well-Being, and SDG 11, which focuses on Sustainable Cities and Communities. Climate resilience and the quality of human settlements are constrained by the state of the urban surface thermal environment. Therefore, improving the urban thermal environment is paramount in shaping the physical context of public health, well-being, and mortality.¹⁰

Rao¹¹ first proposed applying remote sensing data to study the urban surface thermal environment, setting the stage for urban thermal environment research, from the urban canopy and boundary layer to the urban surface layer. Land surface temperature (LST) plays an important role in surface matter and energy exchange; it is considered an important parameter for describing the urban surface thermal environment and explaining its formation mechanisms.^{12,13} The thermal condition index (TCI) is derived from the normalization of the real-time LST using the historical climate range of a specific image, such as the multi-year temperature fluctuation range. Ultimately, it reflects the degree of deviation of the current thermal conditions from the local long-term climate baseline.¹⁴ With the benefit of the consistent global surface observations provided by geostationary satellites, the Copernicus Global Land Service's 5 km/h global LST product covers most Earth's land surfaces between 60°S and 70°N, including hourly LST and 10-day LST daily cycle data.¹⁵ The 10-day LST products provide a complete overview of the daily LST cycle over each 10-day composite, as well as the maximum, median, and minimum LST values for each 10-day period. These compounds can be used to derive the 10-day TCI, with values ranging from 0 to 1. A higher TCI value indicates that the LST during a specific 10-day period is lower compared with the overall historical long time series temperature range.^{14,16}

Typically, the TCI is estimated based on data from polar-orbiting satellites with high spatial resolution using daytime LST or brightness temperature.^{17,18} However, owing to the time-dependent nature of LST, polar-orbiting satellites acquire only the instantaneous surface temperature at a specific observation moment and angle; they cannot reflect the general characteristics of LST over a long time series. Conversely, new-generation geostationary satellites with increased time sampling (hourly resolution) can provide consistent LST data with improved reliability. A 5-km resolution is suitable for macro-scale research, and even the smallest citywide images can provide a view of LST information over successive time periods.¹⁹ Therefore, the Copernicus TCI offers notable advantages for studying the urban thermal environment, especially in large-scale urban thermal investigations. In particular, 10-day LST TCI products eliminate the interference of climate background differences by standardizing the data using local historical baselines. This enables the use of universal evaluation criteria for thermal anomalies, supporting uniform analyses across climatic zones, between cities, and even at global scales.

Traditional case studies have provided insight into the effects of different driving factors on LST in numerous cities. These city-scale samples were often conducted in similar geographic locations or on cities with similar population sizes.^{19,20} For instance, analysis of the relationships between normalized indices for vegetation, water bodies, bare soil, and LST have shown that

impervious surfaces and bare soil can exacerbate increases in LST values, whereas vegetation cover and water bodies can reduce these values.^{21–24} Furthermore, urban expansion, which is associated with human activities and is often measured based on the extent and brightness of nighttime light, results in increased LST values.^{25–27} Liu et al.²⁸ focused on global cities, analyzing the contributions of urban expansion, background climate change, and landscape greening to warming. However, existing research like this that takes global cities as samples are relatively scarce, limiting the development of cooling strategies tailored to different types of cities.

To capture global surface thermal environment heterogeneity and drivers, we used a combination of remote sensing observations and geographically weighted regression (GWR) analysis. Our primary measurements, topography (digital elevation model, DEM), vegetation cover (enhanced vegetation index, EVI), and socioeconomic activity (night lighting, NL), were obtained from remote sensing data. Further, we obtained global Shuttle Radar Topography Mission Global 3 Arc-Second Data Version 003 (SRTMGL3 V003) and Moderate Resolution Imaging Spectroradiometer Vegetation Index 16-Day Composite Product (MOD13A2) from the National Aeronautics and Space Administration, as well as National Polar-orbiting Partnership Visible Infrared Imaging Radiometer Suite (NPP-VIIRS) products that had been corrected for atmospheric effects from the Earth Observation Group, and performed fusion and uniform resolution operations. To avoid a large dimension gap and improve model performance, we defined the normalized quantization indices as DEM_N , EVI_N , and NL_N . Next, we spatially aggregated the quantitative indices for each city by obtaining all the pixels within each vector polygon via partition statistics. Finally, correlation analysis and GWR modeling were performed to explore the global patterns of these drivers of the global urban surface thermal environment.

Filling this gap in the literature has two practical implications. First, LST and TCI mapping can improve understanding regarding surface-atmosphere exchange processes at global and regional scales. This can provide valuable land surface status indicators for surface thermal environment application studies.²⁹ Additionally, by comparing the surface thermal environments of cities and identifying general patterns, more effective mitigation strategies can be developed to ensure climate-specific improvement actions in comparable cities.³⁰ Second, the driving mechanisms of urban surface thermal environments are complex; thus, the analysis of global cities might aid in addressing the limitations of individual case studies and enable specific cities to identify optimal strategies for mitigating the urban effects of climate change.

Cities are hotspots for economic growth and innovation, and the survival of future human populations depends on the climate resilience of these cities. The SDGs advocate for global human action in response to climate change and its impacts. Although the drivers of the urban surface thermal environment are complex and not fully understood, this study provides an integrated approach from a macroscopic perspective. Restricting the drivers to natural and socioeconomic aspects and including both two- and three-dimensional orientations enabled the identification of factors that are directly associated with the urban

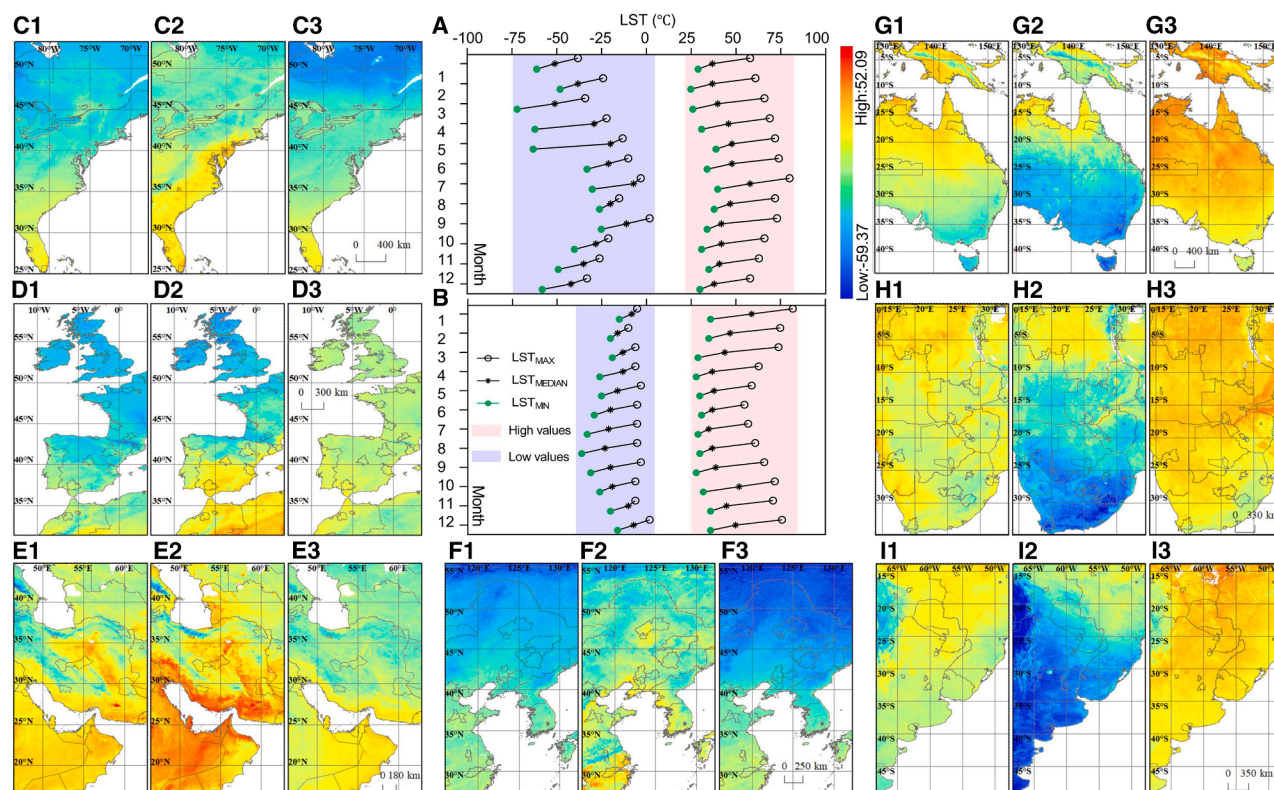


Figure 1. Monthly evolution of the global LST in 2021

(A) and (B) Represent the ranges of LST maxima (LST_{MAX}), minima (LST_{MIN}) and median (LST_{MEDIAN}) in the Northern and Southern hemispheres. This figure contains local magnifications of the LST of North America (C), Europe (D), the Middle East region (E), Northeast Asia (F), Australia (G), Africa (H), and South America (I). Each set of local magnifications includes the LST of April, August, and December, denoted by the numbers 1, 2, and 3 from left to right.

surface thermal environment. Overall, our dataset included 1,469 cities worldwide, distributed in East Asia and the Pacific (EAP), Europe and Central Asia (ECA), Latin America and the Caribbean (LAC), the Middle East and North Africa (MENA), North America (NA), South Asia (SA), and Sub-Saharan Africa (SSA). The dataset included 561 megacities (i.e., with areas greater than 1,000 km²) with different population sizes, 538 medium-sized cities (100–1,000 km²), and 370 smaller cities (with areas less than 100 km²).

The purpose of this study is to explore the characteristics of the urban surface thermal environment and its driving factors from a global perspective. Specifically, our study provides global estimates of the inter-month variability and spatial heterogeneity of the urban surface thermal environment in 2021 based on the latest geostationary satellite data products, using quantitative remote sensing and GIS spatial analysis. Notably, different climatic divisions receive different amounts of solar radiation. Therefore, consistent with the Köppen and Geiger's climate classification world map,³¹ global cities can be grouped according to climate type (hot, arid, warm, and cold climates). Their thermal environment characteristics can then be explored in the context of their climate background (Figure S1). By constructing multivariate datasets, we obtain general patterns of urban thermal responses to topography, vegetation, and nighttime

lighting. In summary, this study attempts to address the limitations of individual case studies and provide data-based support for global urban climate collaborative governance and the realization of the SDGs.

RESULTS

Spatiotemporal heterogeneity of the urban surface thermal environment

Global inter-monthly LST showed variability in the Northern and Southern Hemispheres owing to the movement of the direct solar point (Figure 1). In Northern Hemisphere cities, high LSTs usually occurred from June to August, for example in the eastern Americas, northern Africa, and southern Asia. Conversely, in the Southern Hemisphere, the inter-monthly LST variation curve was concave, and for most Southern Hemisphere cities, LST showed a downward trend from March to August and increased thereafter. The global thermal environment information conveyed by LST is significantly shaped by climatic background factors such as geographical latitude and atmospheric circulation. In contrast, TCI data offer a more standardized perspective of real-time LST against local historical LST baselines. This is conducive to the consistent assessment of the thermal environmental characteristics of global cities. Figure 2 shows the TCI

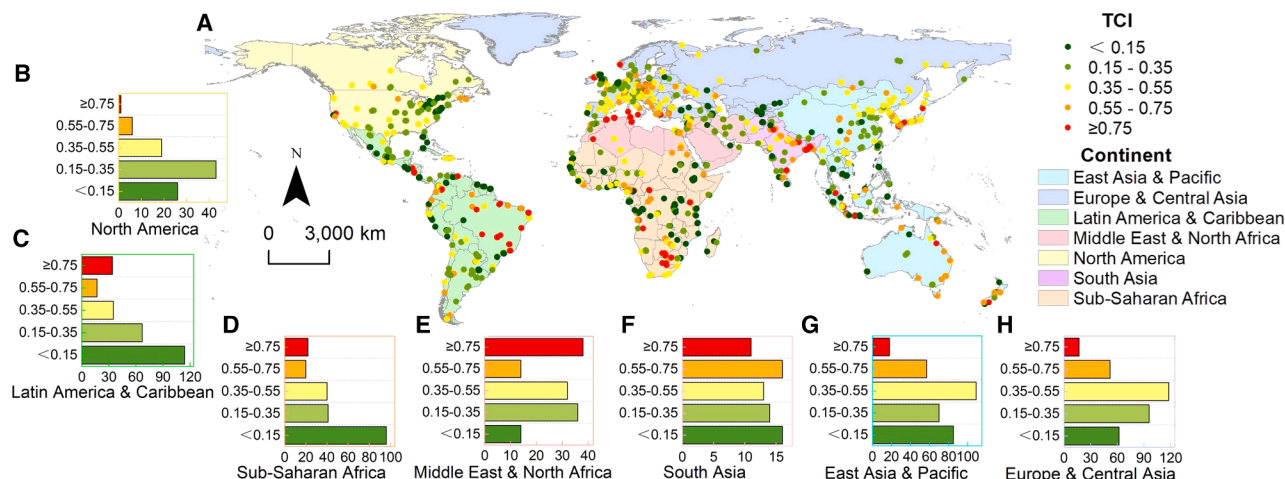


Figure 2. TCI
(A–H) Spatial distribution in global cities (A) and numerical distribution in each continent (B–H).

distribution of 1,469 cities worldwide. The proportion of cities with a TCI exceeding the global average (0.35) was 47% (686/1469). These cities were mainly distributed in the MENA, SA, EAP, and ECA regions, especially along the Mediterranean coast, central Asia, and northeastern Asia. Meanwhile, some cities in the western Americas, the northeastern SA region, southern Africa, and eastern Australia had relatively high TCI values, such as San Mateo (0.62), Cachoeira (0.96), Cacadu (0.54), and Rockhampton (0.75). This indicates that the LST of these cities during the study period was at a historically low level. In contrast, cities with TCI values lower than the global average (accounting for 53%) were mainly distributed in the northeastern NA, SSA, and EAP regions. Among them, the TCI values of 266 cities were less than 0.05, including Lusaka (0.02), Merkez (0.04), and Thrissur (0.01). In particular, in the LAC and SSA regions, there were a significantly greater number of cities with TCI values lower than 0.15 compared to those with high TCI values. The situation in the NA region was also not optimistic. This indicates that the LST of these cities is at a historical high, and their thermal environment conditions are a cause for concern.

Furthermore, combined with the global climate background, the average TCI values from high to low are as follows: warm (0.41) > cold (0.36) > arid (0.32) > hot (0.28). The highest median TCI value (0.41) was obtained for cities in cold regions, followed by those in warm (0.39), arid (0.32), and hot (0.14) regions. This indicates that it is more common for the urban thermal environment in warm and cold regions to be low compared to historical levels. The urban thermal environment in arid regions closely resembles historical levels, whereas the urban temperature in hot areas closely aligns with historical highs. This difference indicates notable variations in the abnormal thermal environment characteristics of different climate zones. TCI effectively demonstrates the unique patterns of LST deviations from historical baselines in various regions, providing a comprehensive assessment framework for the multi-dimensional analysis of the global urban thermal environment.

The spatial heterogeneity of driving variables in the thermal environment

Plains and coastal areas have superior natural conditions that are conducive for urban construction and human settlement, resulting in the global DEM_N values generally approaching 0 (Figures 3 and S2). Analysis shows that the global average DEM_N is 0.098, among which 72% of the cities have DEM_N values lower than the average. This verifies that low-altitude areas accommodate more cities. From the geographical distribution, the altitude of the LAC, SA, and SSA regions is relatively high, and their average DEM_N values were 0.172, 0.141, and 0.136, respectively. There were only 31 cities with an average DEM_N value exceeding 0.5, and these cities were mainly concentrated in the SA and LAC regions.

The differences in the economic development levels of global cities lead to obvious differentiation characteristics in the spatial distribution of nighttime light intensity (Figure S3). The NL_N values of the global cities were generally low, with an average of 0.138. Among them, the proportion of cities with NL_N levels lower than the global average was 67%. The average NL_N values of the MENA, NA, and EAP regions were relatively high (0.221, 0.186, and 0.167, respectively), reflecting the high urbanization and economic activity in these regions. However, the NL_N in SA and SSA cities was generally low, confirming that the cities in these regions lag behind in terms of infrastructure construction, economic scale, and the intensity of human activities at night; thus, their overall development level could be improved.

The numerical distribution of EVI_N in global cities was uniform, and there was obvious spatial heterogeneity (Figure S4). The global average EVI_N was 0.562. The urban vegetation coverage of the ECA and NA regions was relatively high, and their average EVI_N values reached 0.683 and 0.655, respectively. The average EVI_N values of the EAP and LAC regions were 0.592 and 0.582, respectively, which are close to the global average. In sharp contrast, the average EVI_N value of the cities located in the MENA region was only 0.276, significantly lower than that of

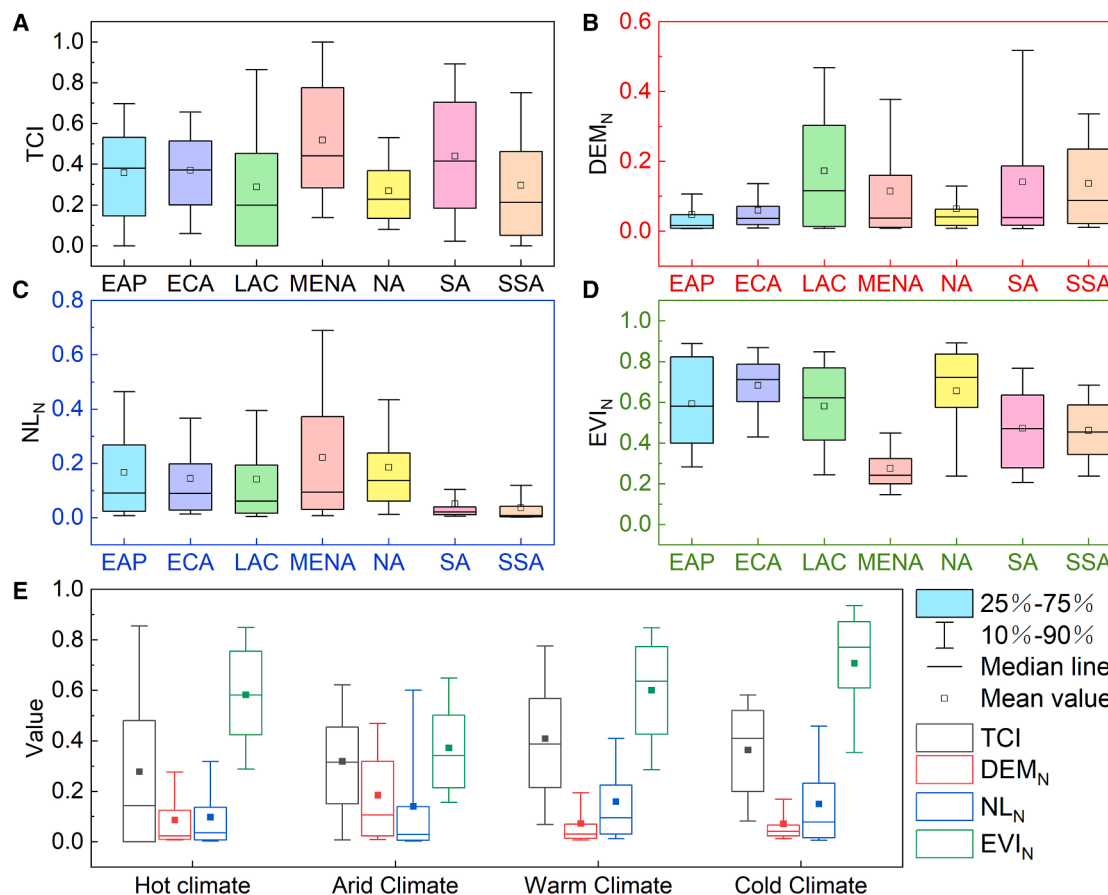


Figure 3. Numerical distribution of DEM_N , NL_N , EVI_N , and TCI in different climate regions

other areas, highlighting the ecological feature of relatively scarce vegetation coverage in this region.

Furthermore, considering the global climate zones, the DEM_N was the highest in arid areas (with an average value of 0.185 and a median value of 0.107). Overall, NL_N and EVI_N were higher in warm and cold regions than in hot and arid regions, which is consistent with the distribution of the TCI in the climate zones. Therefore, the driving mechanism of the TCI requires further exploration.

Response patterns of thermal environments to driving factors

Correlation analysis

Considering the similarity of the climate background and differences in urban development gradients, Spearman correlations between thermal environment influencing variables (DEM_N , NL_N , and EVI_N) and the TCI showed differentiation characteristics (Tables 1 and S2; Figure S5). Specifically, the positive correlation between DEM_N and the TCI was more prominent in the NA ($\rho = 0.481$) and EAP ($\rho = 0.315$) regions. This reflects that the urban TCI in this area increases with increasing DEM , indicating that the heat stress has a tendency to weaken in more elevated terrain. The DEM_N and TCI in the cities in the MENA ($\rho = -0.420$) and SA ($\rho = -0.451$) regions showed a negative correlation, indi-

cating that the low altitude is beneficial to the urban thermal environment. The correlation between NL_N and the TCI showed obvious regional differentiation. For instance, NL_N and the TCI were negatively correlated in the NA ($\rho = -0.280$) and EAP ($\rho = -0.276$) regions, indicating that cities with higher nighttime light intensity (representing human activity and the density of the built environment) tend to face higher heat stress. However, in the MENA region ($\rho = 0.338$), there was a positive correlation, indicating that the higher the intensity of the nighttime light, the lower the heat stress. The cities with a positive correlation between EVI_N and TCI were mainly distributed in the EAP ($\rho = 0.386$), ECA ($\rho = 0.164$), and MENA ($\rho = 0.126$) regions, indicating that vegetation coverage has a positive effect on the improvement of the thermal environment in these regions. However, no significant correlation was observed in the SSA and SA regions, reflecting that the vegetation regulation effect might be limited by local climatic conditions or vegetation types.

GWR analysis

Considering the spatial heterogeneity of the global urban surface thermal environment and its drivers, we performed spatial correlation analysis (Figure S6). DEM_N showed a positive spatial correlation with the TCI in low-altitude areas (such as the NA region, Europe, and the EAP region), and this positive correlation was stronger in the eastern United States, central Europe, the SSA

Table 1. Spearman's correlation coefficient between DEM_N, NL_N, EVI_N, and TCI

Continent variables	TCI & DEM _N	TCI & NL _N	TCI & EVI _N
East Asia & Pacific	0.315**	−0.276**	0.386**
Europe & Central Asia	0.109*	−0.161**	0.164**
Latin America & Caribbean	0.009	−0.090**	0.066*
Middle East & North Africa	−0.420**	0.338**	0.126
North America	0.481**	−0.280**	−0.097
South Asia	−0.451**	0.075	−0.016
Sub-Saharan Africa	0.231**	0.003	−0.185**

*Correlation is significant at the 0.05 level (2-tailed).

**Correlation is significant at the 0.01 level (2-tailed).

region, and eastern Asia. However, in high-altitude areas where establishing metropolitan areas is challenging (such as Central Asia, the SA region, and the Iberian Peninsula), DEM_N showed a significant negative spatial correlation with the TCI. Compared with DEM_N (coefficient range −2.30 to 3.93), the negative spatial correlation between NL_N and the TCI was dominant (the maximum negative coefficient was −2.67), whereas the positive spatial correlation was not obvious (the maximum positive coefficient was 1.42). The spatial correlation between EVI_N and the TCI was relatively weak, with the coefficient ranging from only −1.07 to 0.79. This could illustrate that, from a global perspective, the regulatory capacity of vegetation cover on the thermal environment is weaker than that of topographic and socioeconomic factors, being limited by vegetation types, evapotranspiration efficiency, or regional climate adaptability.

Furthermore, in this study, the number of cities distributed in each climate zone was statistically analyzed based on the range of driving variables and GWR coefficients (Figure 4). The effect of DEM_N on the thermal environment (TCI) was positive in most cities in hot, warm, and cold regions but negative in those in arid regions. EVI_N had a negative effect on the TCI in most cities in arid areas, especially when the EVI_N value was < 0.21. Additionally, cities in arid regions were found to have relatively underdeveloped urban economies (NL_N < 0.02), and for most cities, NL_N was negatively associated with the TCI. This phenomenon was more obvious in warm climates, where the GWR coefficients were concentrated between −0.5 and −0.08. Further, in cold climatic zones, DEM_N and EVI_N generally exerted cooling effects on the urban surface thermal environment, whereas a high NL_N value showed negative effects in this regard (GWR coefficient = −0.08).

DISCUSSION

A global perspective on the response patterns of the urban surface thermal environment has not been well developed. We scrutinized the global distribution of thermal conditions in more than 1,400 cities, providing insight into climate change adaptation and mitigation. Our study presents information that is crucial for effectively understanding existing natural and anthropogenic drivers of the urban surface thermal environment in diverse climate zones. Distinct from the typical LST and TCI data used

in traditional research, this study advances the existing literature by endeavoring to utilize new-generation satellite products to identify the global distribution patterns of the urban surface thermal environment. These new products provide hourly LST data alongside 10-day TCI data that encapsulates the normalized outcomes of historical long-term LST. The results align with the results of previous studies but can be applied on a global scale. Our experiments have demonstrated that the spatial heterogeneity of TCI better reflects variations in the thermal environment of different cities relative to their own baselines. Meanwhile, large-scale thermal environment surveys will aid in providing reliable support for formulating climate adaptation strategies and achieving the SDGs.

The LST data obtained through remote sensing can conveniently and accurately assess the surface thermal environment at different spatiotemporal scales. For instance, existing studies have widely applied LST data derived from polar-orbiting satellite sensors such as Landsat, MODIS, or ECOSTRESS to studying the urban heat island effect, thermal comfort, and thermal risk.^{32–34} These studies have reached the consistent conclusion that rapid urbanization contributes to increased pressure on the urban thermal environment. This study, based on the LST data provided by geostationary satellites, has enriched the understanding of global urban thermodynamics. Our results show that the inter-monthly variation of global LST is relatively stable. A high LST usually occurs from June to August in Northern Hemisphere cities, whereas the LST in most Southern Hemisphere cities shows a downward trend from March to August, followed by an increase. Traditionally, large cities and urban agglomerations with dense populations, high levels of economic activity, and high sensitivity to climate change have received most of the attention in studies on the thermal environment. For instance, in international metropolises such as Los Angeles, Tokyo, Berlin, and Shanghai, the heat island effect influences the quality of life and health status of local residents, as well as indirectly affects the surrounding areas, changing the local climate.^{35–38} However, the significance of small- and medium-sized cities in research on the global thermal environment cannot be ignored. Studies have shown that the phenomenon of encroachment on natural land in the built-up areas of developing cities also increases LST. Such cities can become a micro-heat source in the regional climate system, and their climate adaptability is relatively insufficient.^{39–41} In contrast, our research results indicate that in hot and arid regions with relatively small urban areas, efforts need to be made to improve the management of urban high temperatures. Our dataset contained small- and medium-sized cities from across the world (Figure S1). Among them, the TCI values of most small cities with an area of less than 100 km² were lower than 0.2, and these were mainly distributed in the hot and arid regions near Europe and the equator. The average TCI of medium-sized cities (with an area of 100–600 km²) was 0.34, and they were mainly distributed in the LAC and SSA regions, Europe, and SA. This indicates that, overall, the thermal environmental trends in small- and medium-sized cities are equally worrying.

The results of this study verified the general patterns of the urban surface thermal environment related to topography, vegetation, and socioeconomic activities, and captured the spatial non-stationarity of the influence of these variables on the thermal

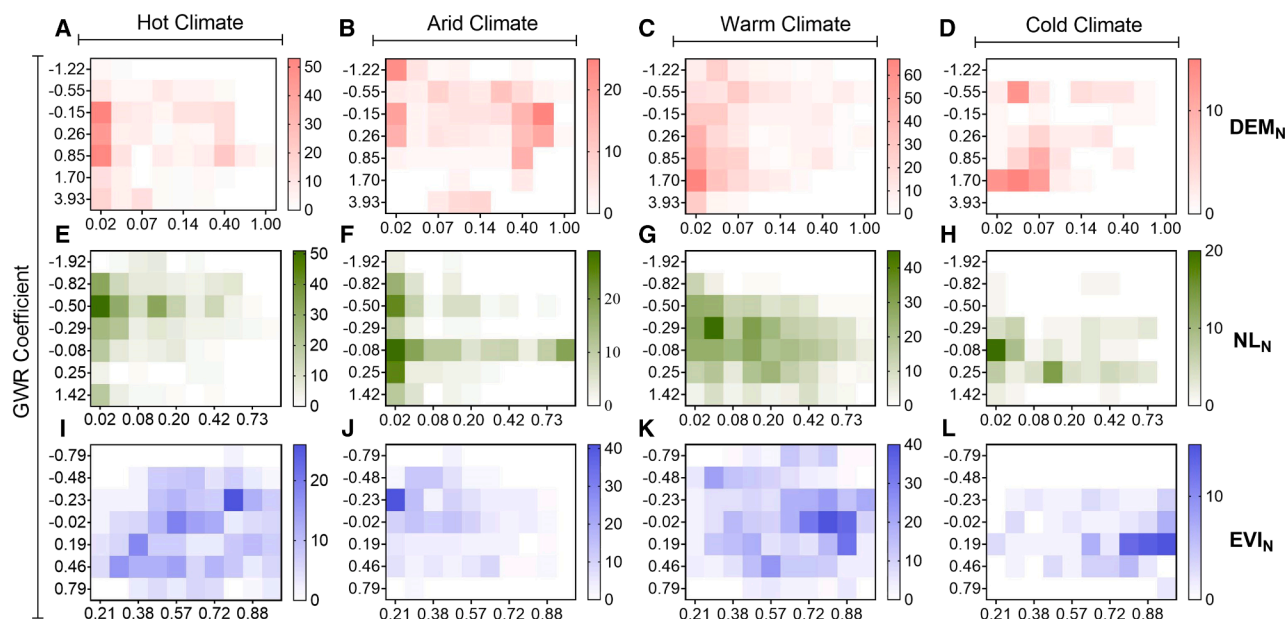


Figure 4. The number of cities categorized by GWR correlation coefficients and driver variable levels

The driving variables and GWR coefficients were classified using the natural breaks method, forming the x axis and y axis of each sub-graph, respectively. The x axes in A–D represent DEM_N levels, those in E–H represent NL_N levels, and those in I–L represent EVI_N levels. Color blocks of each subplot represent the number of cities. The vertical color bar adjacent to each subplot quantitatively represents the city count distribution, with the chromatic gradient transitioning from light to dark hues corresponding to increasing urban numbers. For DEM_N , most cities in hot, warm, and cold regions exhibit positive GWR correlation coefficients with TCI. Especially when DEM_N is below 0.02, the GWR coefficients typically exceed 0.26. In arid regions, there are 23 cities where DEM_N is less than 0.02 and the GWR coefficient between DEM_N and TCI is below -1.22 , and 25 cities where DEM_N ranges between 0.4 and 0.6 and the GWR coefficient falls between -0.55 and -0.15 . For NL_N , the GWR coefficients for most cities in hot and warm regions range from -0.5 to -0.08 , while those for most cities in arid and cold regions range from -0.29 to 0.25 . For EVI_N , the GWR coefficients for 26 cities in hot regions range between -0.48 and -0.23 , with EVI_N values distributed between 0.72 and 0.80. In arid regions, there are 66 cities where EVI_N values are below 0.21, and their GWR coefficients are typically negative. In warm and cold regions, a relatively larger number of cities have EVI_N values exceeding 0.72 and GWR coefficients ranging between -0.23 and 0.19 .

environment. The influence of DEM on the thermal environment has been discussed little in previous case studies. Guo et al.⁴² pointed out that the influence of DEM on changing LST is approximately 13%. Specifically, when the DEM is between 24.4 and 361.2 m, it is negatively correlated with LST. We have noticed that globally, the effect of DEM_N on TCI was stronger in low-altitude areas ($DEM_N < 0.02$) than in high-altitude areas. Second, previous studies such as those conducted in the Great Lakes region of North America, southern Europe, central India, and northern China, have revealed that changes in LST are closely related to socioeconomic activities.^{43,44} For example, LST is usually significantly and positively correlated with gross domestic product, population size, and electricity consumption, which tend to be associated with brighter nighttime lighting.^{45–48} The present study extended these findings and showed that economic activity (represented by NL_N) is also associated with a deteriorated thermal environment. However, in some regions (such as the MENA region, China, United States, New Zealand, and Australia), there was a positive correlation between NL_N and the TCI. That is, the higher the NL_N , the higher the TCI value, and the smaller the heat stress at that time. This indicates that in economically developed situations, cities with higher planning levels, investment in green infrastructure, or energy efficiency measures tend to have a stronger ability to improve the thermal environment.

Finally, for the urban surface, there are two major pathways through which vegetation leads to cooling, namely, shading and evapotranspiration.⁴⁹ Peng et al.⁵⁰ evaluated the relationship between the EVI and LST in 419 major cities around the world. Their results indicated that vegetation plays a key role in reducing LST during the day. Li and Liu⁵¹ also pointed out that increasing urban vegetation coverage may be an effective strategy for mitigating urban heat risk. Consistent with this previous study, the present study identified the positive effects of the EVI on the urban thermal environment based on TCI data, although it was weak from a global perspective. Figure S6 showed that cities with weak EVI_N -TCI correlations were widely distributed across NA, LAC, ECA, and EAP. This is because the scale of the area investigated may impact these positive effects of vegetation.^{52,53} For example, studies have shown that vegetation in arid areas may increase LST, especially at the beginning of the growing season.^{54,55} Vegetation types and combinations can also affect the urban surface thermal environment.^{56,57} Reportedly, the LST index may underestimate the cooling effect of vegetation.⁵⁸ Therefore, the accurate assessment of surface parameters at an acceptable resolution can contribute to better urban planning and improved understanding regarding local climatic conditions. In future studies, the unique role of vegetation should be further investigated to clarify strategies for improving the urban surface thermal environment via nature-based solutions.⁵⁹

Table 2. Explanation table of symbols

Symbols	Explanation	Symbols	Explanation
LST	land surface temperature	GWR	geographically weighted regression
TCI	thermal condition index	EAP	East Asia and Pacific
DEM	digital elevation model	ECA	Europe and Central Asia
DEM _N	normalized DEM	LAC	Latin America and Caribbean
NL	night lighting	MENA	Middle East and North Africa
NL _N	normalized NL	NA	North America
EVI	enhanced vegetation index	SA	South Asia
EVI _N	normalized EVI	SSA	Sub-Saharan Africa
ρ	Spearman's correlation coefficient	–	–

With increasing urbanization and population sizes, socioeconomic development and human activities are key to future increases in urban heat. In this context, the SDGs advocate for strengthening resilience and adaptation to climate-related and natural disasters in all countries, while integrating climate change measures into national policies, strategies, and planning.⁶⁰ The spatial optimization of urban areas will be crucial for improving the thermal environment, which has significant implications for global sustainable development. Therefore, continuing to neglect the issue of the urban surface thermal environment will adversely impact the future development and livability of cities.

Mayors seeking to enhance the resilience of their cities to warming climates and extreme heat events can gain valuable insight from this study. For example, our analysis shows that vegetation in cities is effective in mitigating LST increases, aligning with the findings of case studies based on high-resolution data.^{61,62} However, our results additionally enable practitioners to adequately reflect local conditions. As another example, economic activity, expressed as a measure of nighttime lighting, reduces thermal comfort in most areas, implying that anthropogenic activities can have a negative impact on the thermal environment. City-scale studies have suggested that the built environment is a direct driver of heat retention in cities.^{63,64} Therefore, policies aimed at mitigating the impacts of anthropogenic activities, e.g., reducing energy consumption and cutting exhaust emissions from combustion engines—or especially by unsealing surfaces and providing vertical greening—can contribute to enhancing the urban surface thermal environment. Future research should combine multi-source, socioeconomic indicators (such as gross domestic product, population, impervious surface area, and building data) to provide a comprehensive analysis of the relationship between human activities and the urban surface thermal environment from a global perspective.

In conclusion, the findings of this study enhance the current understanding of the urban surface thermal environment on a global scale. These findings can serve as a valuable resource

for city officials and planners, assisting them in making informed decisions regarding strategies to enhance the resilience of their cities to warming climates and extreme heat events, which are crucial for global sustainable development.

METHODS

Data and sources

In this study, we expanded the application of multi-source remote sensing data from a global perspective. To ensure data consistency, we spliced and fused a large amount of data, extracted and cropped them, and re-projected them into a unified WGS84 coordinate system for visual display. Further, the extraction of the remote sensing data was based on urban boundaries using zoning statistics to ensure that they would match. Table S1 contains all the data and preprocessing procedures employed in this study. The data included vector boundary and remote sensing image data as well as urban surface thermal environment data derived from the 10-day LST TCI dataset provided by a series of geostationary satellites. DEM data were derived from SRTMGL3 V003 products, the impact of vegetation (EVI) was determined based on MOD13A2 16-day synthetic products, and socioeconomic activity (NL) data were obtained from NPP-VIIRS monthly NL data.

This study used TCI products starting from the 11th of each month in 2021; thus, the composite images included data from the 11th to the 20th of each month. The parameters included the maximum, minimum, and median LST, the TCI, and the effective pixels of LST. To investigate the distribution patterns of the global urban thermal environment, this study fused month-by-month images to reflect the annual level, assigned raster attributes to urban elements, and aggregated global TCI data through zonal statistics. The TCI was calculated according to Equation 1 as follows:

$$TCI = \frac{[LST_{Max}]_{Max} - LST_{10DayMax}}{[LST_{Max}]_{Max} - [LST_{Max}]_{Min}} \quad (\text{Equation 1})$$

The daily LST maximum (LST_{Max}) usually occurs between local noon minus 1 h and local noon plus 2 h. $LST_{10DayMax}$ represents the median LST_{Max} observed during the 10-day compositing period. $[LST_{Max}]_{Max}$ and $[LST_{Max}]_{Min}$ represent the upper and lower LST_{Max} values observed over a 30-year period (1985–2015).

The TCI values represent the normalized LST, with a range of 0–1. A high TCI indicates that the heat stress during a specific 10-day period was low compared with the historical LST range. As LST is strongly controlled by energy fluxes, the 10-day TCI indirectly reflects surface moisture through near-surface radiation and aerodynamic conditions. Therefore, TCI values can effectively reflect thermal environmental information concerning the state of the surface energy balance.

Data normalization

Owing to the large differences in the ranges and units of the different datasets, standardization was necessary to avoid biases and thus improve comparability between the different indicators. In this study, the minimum-maximum normalization method was used, as shown in Equation 2 below.

$$y = \frac{y_i - y_{\min}}{y_{\max} - y_{\min}}, \quad (\text{Equation 2})$$

where y represents the normalized value of the variable, y_i denotes the original value of the variable (e.g., original DEM, EVI, and NL), and $y_{\min}$ and $y_{\max}$ denote the minimum and maximum values of the original values, respectively. For each index, the raw data were normalized within the range [0,1], which is suitable for comparative analysis. The normalized values were recorded as DEM_N , EVI_N , and NL_N .

Statistical analysis

Spearman's correlation analysis was performed to determine the relationship between each driving factor and the urban surface thermal environment. The correlation coefficients (ρ values) were derived according to Equation 3 as follows:

$$\rho = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}, \quad (\text{Equation 3})$$

where x represents the TCI value, y represents the value of the driving variables, and $\bar{x}$ and $\bar{y}$ denote the mean values of x and y , respectively. If the driving variable has a positive correlation ($\rho > 0$) with TCI, it means that the larger the variable value, the larger the TCI value. This indicates that the $LST_{10DayMax}$ at that time was historically low and the heat stress was relatively small. In contrast, if $\rho < 0$, it indicates that the variable can increase local heat stress.

Owing to the spatial heterogeneity of the urban surface thermal environment and its drivers, it was necessary to consider spatial location while exploring the mode of action of the different drivers. Thus, we performed GWR analysis using spatial statistics tools in ArcGIS 10.7, in accordance with a Gaussian kernel (adaptive) approach to analyzing spatially assigned proximate number functions of the environment. GWR models, which use locally weighted least squares for point-by-point parameter estimation, have been employed to explore the spatial variability of variables and drivers. They are capable of considering the local effects of spatial objects with high accuracy. Thus, GWR analysis was performed, with the TCI as the dependent variable and DEM_N , EVI_N , and NL_N as the independent variables, using the following equation:

$$y_i = \beta_0(u_i, v_i) + \sum_{k=1}^n \beta_k(u_i, v_i) x_{ik} + \varepsilon_i, \quad (\text{Equation 4})$$

where (u_i, v_i) represents the coordinates of the i^{th} point, $\beta_k(u_i, v_i)$ represents the k^{th} regression parameter at the i^{th} sampling point and is a function of the geographic location, and ε_i represents the random error of the i^{th} region. Between variables, a variance inflation factor of less than 7.5 indicated that the GWR was valid.

The symbols and their interpretations that appear in this study are shown in Table 2. Further details regarding the methods can be found in the supplemental methods.

RESOURCE AVAILABILITY

Lead contact

Requests or questions related to data, methods, or results can be made to the lead contact, Jun Yang (yangjun8@mail.neu.edu.cn).

Materials availability

This study did not generate new unique materials or code.

Data and code availability

The raw and processed data that support the findings of this study are openly available from Figshare at <https://doi.org/10.6084/m9.figshare.24113460.v2>.

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AUTHOR CONTRIBUTIONS

Conceived and designed the experiments, J.Y.; conceptualization, J.R. and F.C.; contributed materials/analysis tools, J.C.X. and Q.G.; data analysis, J.R. and B.Z.; methodology, J.R., F.C., and W.S.; project administration, J.Y.; resources, X.X. and Q.G.; supervision, X.X. and Q.G.; visualization, B.Z. and Q.G.; writing—original draft, J.Y., J.R., and B.Z.; writing—review and editing, F.C., W.S., X.X., J.C.X., and Q.G.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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